

Review

Transforming agriculture with Machine Learning, Deep Learning, and IoT: perspectives from Ethiopia—challenges and opportunities

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Received: 4 June 2024 / Accepted: 26 August 2024

Published online: 01 October 2024

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Abstract

Agriculture holds a crucial position in maintaining livelihoods and securing food sources, particularly in nations such as Ethiopia, where a substantial portion of the population depends on agricultural pursuits. However, meeting the growing demand for food production amidst population growth presents considerable challenges. Recent advancements in technology, particularly in the areas of Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT) offer promising solutions to address these challenges. This paper explores the potential of integrating ML, DL, and IoT technologies in agriculture to revolutionize the sector. By harnessing data-driven insights, farmers can make informed decisions regarding crop management, soil health, and weather patterns, leading to optimized resource allocation and increased productivity. Moreover, IoT devices enable the real-time monitoring and control of agricultural operations, enhancing sustainability and productivity. Despite the opportunities presented by these technologies, there are also challenges to overcome, such as data quality, connectivity issues, and the need for farmer education. However, with concerted efforts and investment, Ethiopia and other agricultural regions can unlock the full potential of ML, DL, and IoT technologies to ensure food security, alleviate poverty, and drive economic development. This review paper offers perspectives on the present status, challenges, and future possibilities regarding the integration of ML, DL, and IoT in agriculture. It underscores the transformative potential of these technologies within the sector.

Keywords Transforming agriculture · Internet of things · Machine Learning · Deep Learning · Decision support systems

1 Introduction

Agriculture is a crucial sector that sustains human civilization, providing essential resources such as food, fiber, and fuel. It plays a pivotal role in addressing global challenges, encompassing poverty alleviation, ensuring food security, and fostering economic development [1]. As the world's population continues to grow, reaching an estimated 9.7 billion by 2050, the pressure on agriculture to meet the increasing demand for food rises significantly [2, 3]. In the context of Africa, agriculture is even more critical, as it remains the primary source of income and employment for the majority of the population [4]. Ethiopia, one of the largest countries in Africa, heavily relies on agriculture to sustain the livelihoods of its people. The sector accounts for about 34% of the GDP and employs approximately 66% of the workforce,

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predominantly characterized by smallholder farmers who practice rain-fed agriculture. This reliance on rainfall makes the sector particularly vulnerable to climate variability and change [5–7]. Additionally, Ethiopian agriculture faces systemic challenges such as limited access to modern technologies, inadequate infrastructure, fragmented land holdings, low productivity, and market access issues [7]. These challenges are further compounded by frequent droughts, soil degradation, and water scarcity.

The global agricultural sector confronts various challenges that impact its productivity and sustainability. Climate change is one of the most pressing issues, with its adverse effects on weather patterns, extreme events, and temperature changes [8, 9]. Depleting natural resources, including water scarcity and soil degradation, further exacerbates the challenges faced by farmers [10]. Food waste and loss in the supply chain contribute to inefficiencies, resulting in decreased food availability and increased food insecurity [11]. Market access challenges limit the ability of farmers, especially smallholders, to reach consumers effectively and earn fair prices for their products [11].

In Africa, particularly in Ethiopia, these global challenges are amplified by region-specific factors. Limited infrastructure, including poor road networks and inadequate storage facilities, hinders the efficient distribution and trading of agricultural products [12]. Subsistence farming, characterized by low technology adoption and limited market orientation, leads to low productivity and vulnerability to external shocks [13, 14]. Water scarcity, a significant concern in parts of Africa, demands innovative solutions for sustainable water management and irrigation [15]. Lack of access to modern agricultural technologies and the lack of knowledge in practices that can sustainably increase agricultural production limit productivity gains and income opportunities for farmers [16–18]. Moreover, poverty and food insecurity persist, necessitating urgent action to ensure sustainable and inclusive agricultural development [19].

As depicted in Fig. 1, agricultural revolution encompasses four stages: traditional agriculture based on manual and animal labor; mechanized agriculture featuring equipment innovations; automated agriculture propelled by accelerated progression; and smart agriculture spearheaded by emerging technologies [20]. By embracing technological integration within intelligent farming practices, this approach fosters long-term viability through the application of Information and Communication Technologies (ICTs) within the farm management ecosystem, leveraging tools such as the IoT, ML, DL, cloud computing, and robotics [21–23].

The integration of ML, DL, and the IoT in agriculture offers transformative solutions to address the sector's challenges [24–29]. Harnessing data-driven insights, ML and DL algorithms equip farmers with the capability to make informed decisions utilizing real-time data concerning their crops, soil conditions, and weather patterns. By scrutinizing extensive agricultural data gathered through IoT devices, these technologies facilitate precise and optimized resource utilization, such as deploying precision irrigation systems to minimize water wastage and enhance crop vitality [25, 30]. ML and DL models significantly contribute to refining decision-making procedures by uncovering intricate patterns and associations that might elude human decision-makers. As a result, farmers can optimize their farming practices, select suitable fertilizers and pesticides, and receive personalized recommendations tailored to their unique circumstances [31]. By

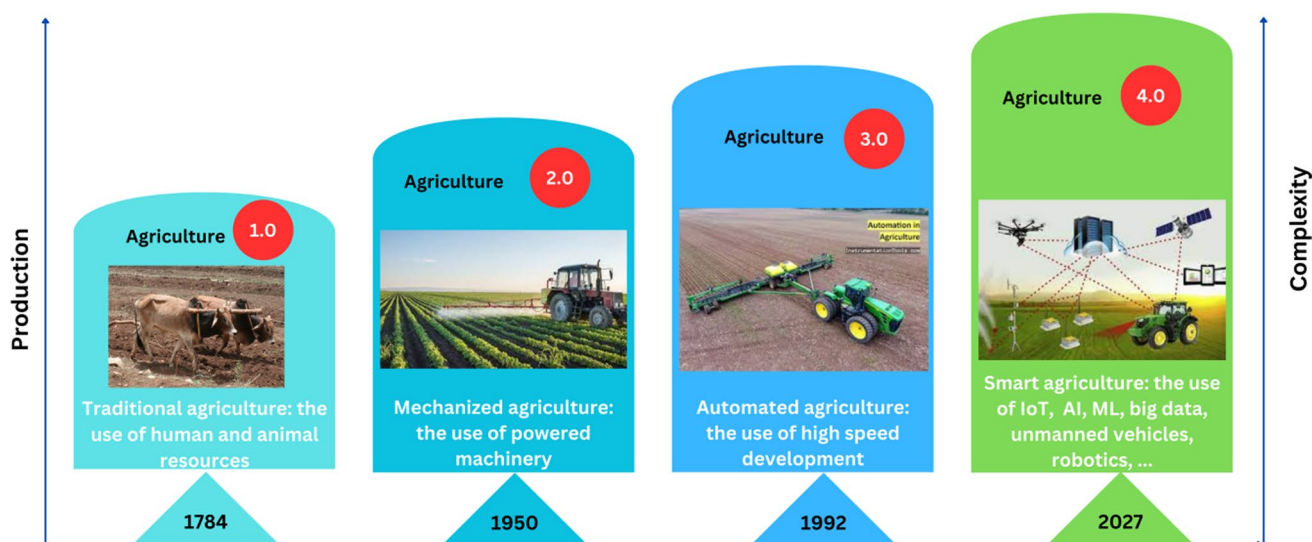


Fig. 1 Traditional, mechanized, automated, and smart agriculture are the four agricultural revolutions indicated from left to right

harnessing ML, DL, and IoT technologies, farmers can enhance overall agricultural productivity, detect diseases and pests early, and optimize crop yield prediction, leading to sustainable and efficient agricultural development (Fig. 2).

Various studies have explored the applications of ML, DL, and the IoT in agriculture. Pangarkar et al. [32] demonstrated the significance of crop yield prediction for various stakeholders such as traders, agri-businesses, and government agencies. While they initially utilized statistical and crop simulation methods, their study explored the potential of ML models as promising alternatives. They assessed the predictive performance of various ML models for forecasting Cluster bean yields in the Bikaner district of Rajasthan. Notably, Support Vector Regression with a linear kernel emerged as the most effective model, boasting the highest R^2 (98.31%) values, thus rendering it to be suitable for yield forecasting due to its robust sample and global validity.

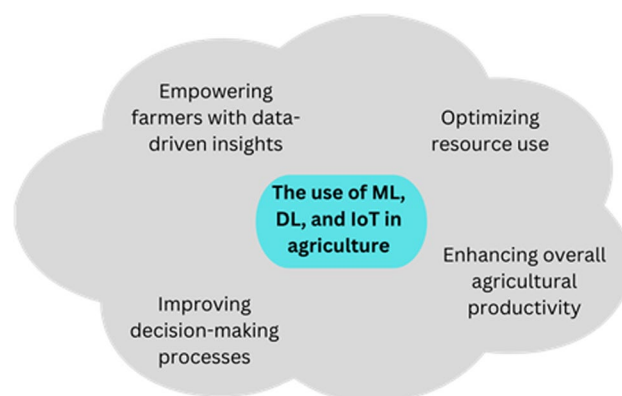
Liu and Wang [33] used DL for Disease and Pest Detection, offering proactive pest management solutions. Ikram et al. [34] introduced an IoT device enhanced with ML capabilities, which integrates various soil sensors and algorithms to monitor soil conditions and provide tailored guidance for different crops. This device effectively enhanced crop yields and optimized resource management, showcasing the potential of technology in agricultural advancements. Obaideen et al. [35] implemented IoT-based smart irrigation systems for the optimization of irrigation, conserving water in water-scarce regions. Akilan and Baalamurugan [36] address challenges in India's agricultural sector by developing an IoT and DL-based system. It automates weather forecasting and field monitoring, enhancing crop productivity and disease prediction. The system achieves high accuracy, with 94% for weather forecasting and 98% for field monitoring, supporting farmers in increasing crop production efficiently. Kumar [37] highlights the significance of livestock management in agriculture, proposing a layered framework integrating IoT and DL for real-time monitoring. The framework involves sensor-rich devices for data collection, data processing for outlier rejection, and DL analysis for detecting animal health issues. Experimental results suggest cost-effective production enhancement, with future research avenues discussed. Purandare et al. [38] applied ML and IoT for harvesting and post-harvesting management, reducing food waste and enhancing supply chain efficiency.

In the context of Ethiopian agriculture, the integration of IoT, ML, and DL technologies presents significant opportunities. IoT applications include precision farming, irrigation monitoring, and livestock management, which can enhance crop performance, boost farm productivity, and reduce harmful pesticide use. IoT devices such as RFID, wireless sensor networks, and GPS facilitate automated monitoring and management of agricultural processes [39]. The synergistic application of ML, DL and IoT addresses various farming challenges and improves prediction accuracy for agricultural applications such as soil characteristics, yield prediction, crop rotation and monitoring disease and pest detection and management.

Ayalew and Lohani [40] combined support vector machines with DL algorithms to predict crop yields in Ethiopia, achieving 96% accuracy. Amtate and Teferi [41] developed a CNN model for classifying coffee beans, achieving 99.89% accuracy across nine classes from six different regions. In cotton farming, Azath et al. [42] created a CNN model to detect leaf diseases and pests, reaching 96.4% accuracy. Similarly, Gayatri et al. [43] applied DL techniques to hyperspectral images for cotton leaf disease detection. These studies illustrate the potential of Deep Learning to enhance agricultural practices in Ethiopia, from crop yield prediction to disease detection and product classification, offering promising solutions for improving productivity and quality in the sector.

This review paper aims to analyze the existing literature on the use of ML, DL, and IoT in agriculture, with a specific focus on the distinct conditions and obstacles encountered in Ethiopia. The research question guiding this review is: how can ML, DL, and IoT technologies be effectively harnessed to address the unique challenges and enhance the productivity

Fig. 2 Application of ML, DL, and IoT in the agricultural sector



and sustainability of Ethiopian agriculture? By addressing this question, the paper will explore various aspects of agriculture, including crop monitoring and yield prediction, disease and pest detection and management, soil and nutrient management solutions, optimization of irrigation and water management, weather forecasting and climate modeling, farm management and decision support systems, livestock management with ML, DL, and IoT, and harvesting and post-harvesting management solutions.

By critically analyzing these technologies' successes, limitations, and possibilities, this review seeks to provide valuable insights into how they can be effectively harnessed to revolutionize agriculture and foster sustainable development. The value of this study lies in its potential to guide policymakers, academics, and other stakeholders in developing tailored approaches and initiatives to address the unique challenges faced by the agricultural sector. Furthermore, the review will explore the importance of involving local stakeholders in policy formulation and technology deployment, providing concrete policy recommendations for sustainable and inclusive agricultural practices. Emphasizing the regulatory frameworks and the need for stakeholder involvement will strengthen the practical implications of this study.

Additionally, the review will consider the implications of emerging technologies and their potential applications in Ethiopian agriculture, highlighting opportunities for innovation and advancement. By implementing technology-based solutions, this review aims to promote sustainable and equitable agricultural practices throughout the country, ultimately contributing to enhanced food security, reduced poverty levels, and stimulated economic expansion.

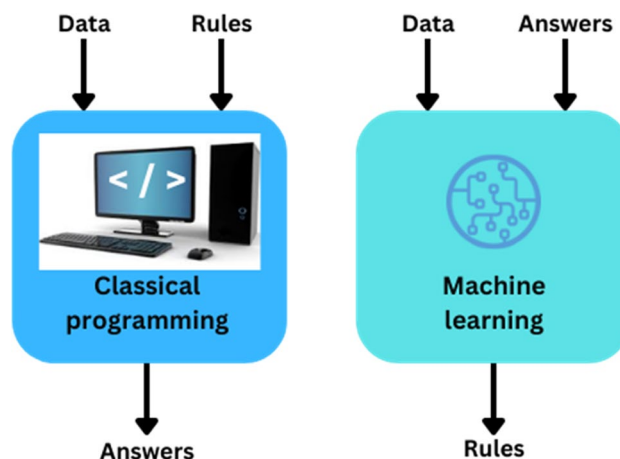
2 Machine Learning, Deep Learning, and IoT: empowering agricultural innovation

2.1 An overview of ML in agriculture

ML falls under the umbrella of artificial intelligence (AI), enabling computer systems to learn from data and autonomously make predictions or decisions without the need for explicit programming [44]. Unlike classical programming, where computers are explicitly instructed to perform specific tasks using predefined rules and logic, ML algorithms detect intricate patterns and relationships in data, enabling autonomous decision-making and predictions (Fig. 3). ML excels in handling complex and evolving situations, making it suitable for tasks where rules are unclear or subject to change. In the realm of agriculture, ML has garnered considerable attention owing to its capacity to analyze extensive datasets sourced from diverse outlets like satellite imagery, weather data, and soil samples [45]. This capability enables ML to furnish valuable insights and enhance farming practices [46].

ML encompasses a range of types, including supervised, unsupervised, and reinforcement learning. Supervised learning, a fundamental facet of ML, entails training a model on a labeled dataset to forecast outcomes for new data [47]. This method finds application across diverse fields, such as agriculture for predicting crop yield [47] and genetics for forecasting complex traits and disease phenotypes [48]. Musicant et al. [49] introduced a unique challenge in supervised learning, wherein training data comprises aggregate outputs, proposing adaptations of k-nearest neighbors, neural networks, and support vector machines for this context. Erdoğan et al. [50] expanded on supervised training by introducing a framework for leveraging partially labeled data to enhance classification performance. Collectively, these studies underscore the adaptability and promise of supervised learning across various domains.

Fig. 3 In classical programming (left), Data and Rules are taken as input to provide Answers. In contrast, ML (right) takes Data and Answers as input and produces Rules as output



Unsupervised learning, on the other hand, involves training ML algorithms on unlabeled data to uncover inherent patterns or structures within the dataset independently [51]. This type of learning is beneficial for tasks like clustering similar crops or detecting anomalies in agricultural data [52]. Reinforcement learning entails training a ML model to interact with an environment and learn from feedback, typically in the form of rewards or penalties. In agriculture, reinforcement learning can be utilized to optimize irrigation schedules, where the model learns to make decisions based on feedback regarding crop health and water usage. Saikai et al. [53] showcase the potential of reinforcement learning (RL) in optimizing irrigation schedules for agriculture. They focus on wheat production and water-efficient irrigation, respectively. Yang et al. [54] corroborates these findings, indicating that deep RL surpasses traditional and ML methods in irrigation scheduling. Additionally, Chen et al. [55] proposes a deep Q-learning approach for rice irrigation decision-making, emphasizing water conservation without yield compromise. These studies collectively underscore RL's capacity to enhance irrigation efficiency and crop yield.

ML algorithms, including decision trees, support vector machines, random forests, neural networks, and ensemble methods, have found extensive application across different sectors, including agriculture, energy, healthcare, transportation, manufacturing, smart homes, and cities [56–60].

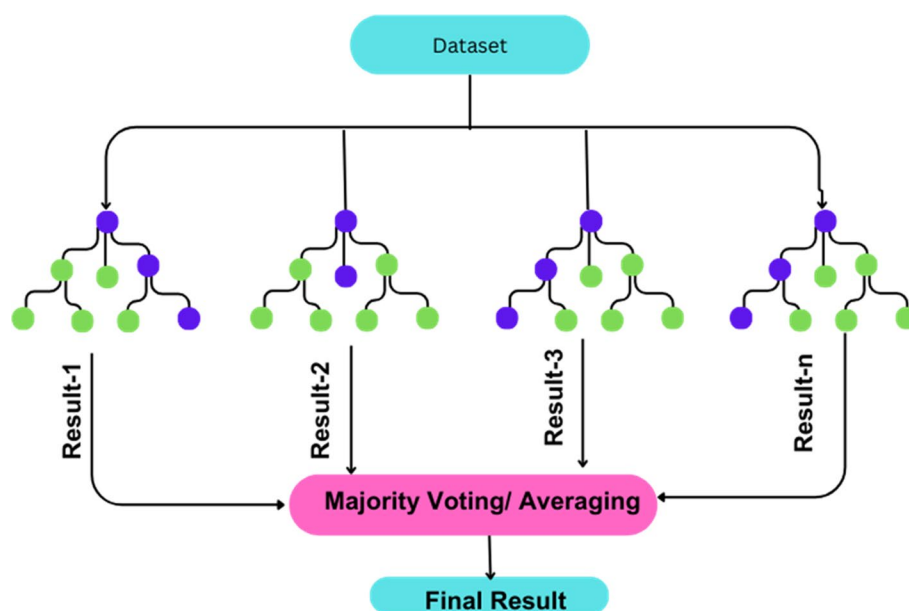
2.1.1 Decision Trees (DTs)

DTs are powerful ML algorithms widely used in agriculture for both classification and regression tasks [61]. These algorithms offer a clear and comprehensible approach to making data-informed decisions by iteratively dividing the data according to the most influential features, thereby constructing a tree-like structure. Each internal node of the tree signifies a decision based on a distinct feature, resulting in diverse branches representing various feature values. The leaf nodes of the tree represent the final predictions or classifications. In agricultural applications, DTs are particularly valuable for crop classification, where they can identify different crops based on plant characteristics, weather conditions, and soil properties [61]. They are also used in disease and pest detection, enabling farmers to identify and classify crop diseases and pest infestations based on symptom patterns and environmental factors. Moreover, DTs find application in predicting crop yields, where they consider historical yield data and factors influencing crop growth, such as weather conditions, soil nutrients, and irrigation practices. DTs provide clear and actionable guidance for farmers, empowering them to make informed decisions regarding resource allocation, crop management, and pest control. However, these structures may encounter overfitting problems, particularly if they become overly deep or complex, and they might have difficulty handling intricate relationships between attributes and the target variable [61]. Multiple decision tree models, like Iterative Dichotomiser 3 (ID3), Chi-square Automatic Interaction Detector (CHAID), and Classification and Regression Trees (CART), have undergone development to enhance their accuracy and effectiveness [62]. These models boast readability, robustness, and practical applicability [63]. Their capacity to manage uncertain data renders them apt for emerging agricultural applications [64].

2.1.2 Random Forest (RF)

RF stands as a robust ensemble learning technique widely applicable in agriculture for both classification and regression purposes. It expands upon the concept of DTs by constructing multiple trees on random subsets of the data and features. Each tree in the forest is trained independently on bootstrapped subsets of the training data, with only a random subset of features considered during each split (Fig. 4). The final prediction in RF is determined by aggregating the predictions of individual trees, typically through methods like majority voting for classification tasks or averaging for regression tasks [65]. In agriculture, RF is utilized across various applications, including crop yield prediction, disease and pest identification, soil property analysis, and precision agriculture. It excels at yield forecasting by considering historical yield data, weather conditions, soil properties, and management practices [66, 67]. Devi et al. [68] emphasize the efficacy of RF in predicting soil fertility, grading, and detecting crop diseases, achieving an impressive 97.2% accuracy in disease detection. Its capability to analyze various factors such as weather conditions and symptoms proves particularly advantageous in these contexts. Sonobe et al. [66] and Ok et al. [69] further showcase the algorithm's potential in classifying crop types, with Sonobe et al. [66] demonstrating superior performance compared to alternative classification techniques. These investigations collectively underscore the adaptability and efficiency of RF in tackling diverse agricultural challenges.

Fig. 4 Workflow of RF prediction: individual trees are trained independently, the final prediction is made by aggregating individual tree predictions



2.1.3 Support Vector Machines (SVM)

SVM represents a potent ML approach with broad-ranging applications, notably in agriculture [70]. Their prowess lies in tasks involving classification and regression, especially when confronted with intricate decision boundaries and datasets with high dimensions [71]. SVMs operate on the principle of minimizing structural risk, enabling them to generalize effectively to novel data and manage datasets containing noise. This is achieved through the establishment of a maximum-margin hyperplane within a transformed input space, which effectively segregates distinct classes while maximizing the proximity to the nearest examples [72]. SVMs have proven effective in various classification challenges, including the examination and modeling of spatially distributed information [71, 73].

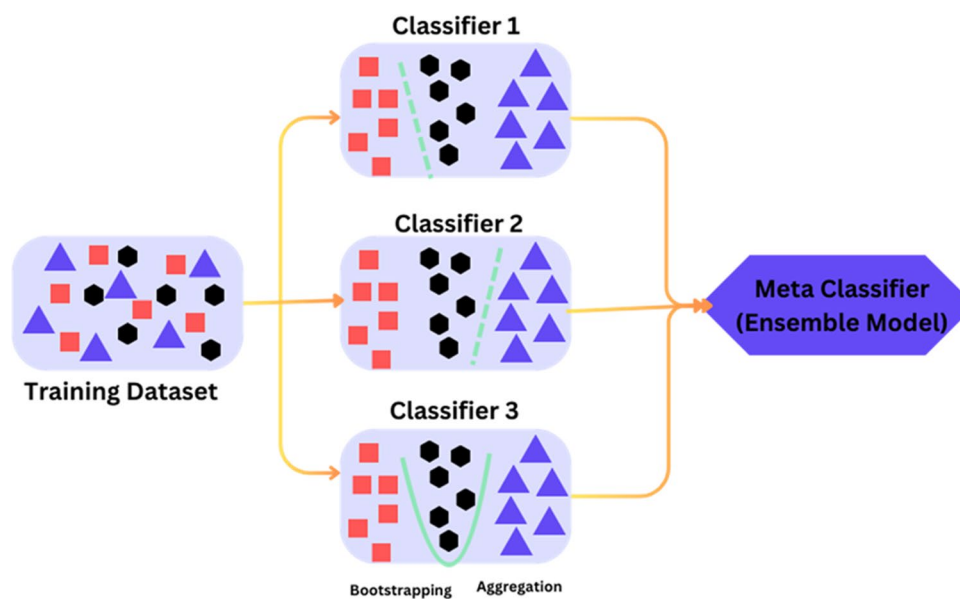
In the field of agriculture, SVM are widely utilized, especially for tasks like categorizing crops, detecting pests and diseases, mapping land use, and predicting yields. Research conducted by Pal and Mather [74] and Shi et al. [75] has emphasized the superior performance of SVM compared to other algorithms in terms of classification accuracy. Pal and Mather [74] specifically highlighted its efficacy with limited training data and high-dimensional datasets. However, challenges such as computational complexity and sensitivity to hyperparameter selection, as pointed out by Shi et al. [75], have been recognized. Despite these challenges, SVM remains a valuable tool in precision agriculture and agricultural research, providing robust solutions for data-driven decision-making.

2.1.4 Ensemble methods

Ensemble methods are also a powerful set of ML models widely used in agriculture and other domains to improve predictive performance and model robustness [76]. These methods hinge on amalgamating multiple models to yield more precise and dependable predictions (Fig. 5). In agriculture, characterized by complex and noisy data, ensemble methods confer notable advantages. Two prevalent ensemble methods include Bagging and Boosting [77, 78].

Bagging entails training numerous models independently on distinct subsets of the training data, created through random sampling with replacement (bootstrapping) [79]. Each model in the ensemble subsequently renders predictions autonomously, with the final prediction derived through majority voting for classification tasks or averaging for regression tasks [80]. A notable algorithm employing Bagging is RF, which amalgamates multiple DTs to attain heightened predictive accuracy and generalization. In contrast, Boosting revolves around constructing a sequence of models, where each successive model rectifies the errors of its precursor. During training, data points that were misclassified by the previous model are assigned higher weights, allowing the next model to focus on those challenging instances. Boosting algorithms, such as Gradient Boosting Machines (GBM) and Adaptive Boosting (AdaBoost), iteratively build models and combine their predictions [79]. Boosting is particularly effective when weak learners are combined to create a strong predictive model.

Fig. 5 Overview of ensemble methods prediction scheme that combine predictions from multiple models to enhance overall accuracy and robustness



In agriculture, ensemble methods find diverse applications. For example, in predicting crop yields, techniques such as RF and GBM are employed to incorporate numerous factors including weather conditions, soil properties, and management practices, resulting in more precise and resilient yield predictions [81–83]. Similarly, in disease and pest detection, ensemble methods amalgamate predictions from multiple models to bolster accuracy and facilitate prompt interventions. Kondaveeti et al. [84] showcased the efficacy of ensemble learning in detecting plant diseases and pests, achieving impressive accuracy rates of 97.8% and 97.56% respectively. Likewise, Ray and Reich [85] also discovered that ensemble techniques offer enhanced predictability of infectious disease patterns, delivering consistent results across different seasons.

Additionally, in precision agriculture, ensemble methods optimize resource allocation by combining predictions from various sensor data sources, helping farmers make data-driven decisions for efficient farm management [86, 87]. While ensemble methods offer numerous advantages, such as improved predictive accuracy and robustness, they can be computationally intensive and require more resources than individual models. Moreover, the final ensemble may be less interpretable than individual models due to the combination of multiple models [88]. Despite these constraints, ensemble methods persist as a valuable and potent strategy in agriculture, furnishing farmers with data-driven insights and bolstering sustainable and efficient agricultural practices.

2.2 DL algorithms for agricultural applications

DL algorithms have been increasingly employed in agricultural applications, including crop classification, disease detection, and yield prediction. Some commonly used DL algorithms in agriculture include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTMs), Generative Adversarial Networks (GANs), Autoencoders, Deep Belief Networks (DBNs), and Capsule Networks [89].

2.2.1 Convolutional Neural Networks (CNNs)

CNNs are a fundamental DL algorithm widely applied in agricultural settings due to their efficiency in extracting statistical patterns from large-scale and high-dimensional datasets, especially those involving visual inputs such as images (Fig. 6) [90]. Key aspects of CNNs relevant to agricultural applications include: (1) CNNs excel at identifying local patterns and combining them into more complex representations, enabling effective image recognition, segmentation, and classification tasks [91], (2) by stacking multiple layers, CNNs create a hierarchy of abstracted features, allowing them to recognize objects at various scales and levels of detail [92, 93], (3) CNNs employ spatial convolutions and shared weight parameters across the input space, resulting in reduced memory requirements and faster processing times [94, 95].

In agricultural applications, CNNs have demonstrated success in areas such as crop classification, disease detection, and weed identification [96–98]. Boulent et al. [96] highlights the remarkable efficacy of CNNs in identifying crop

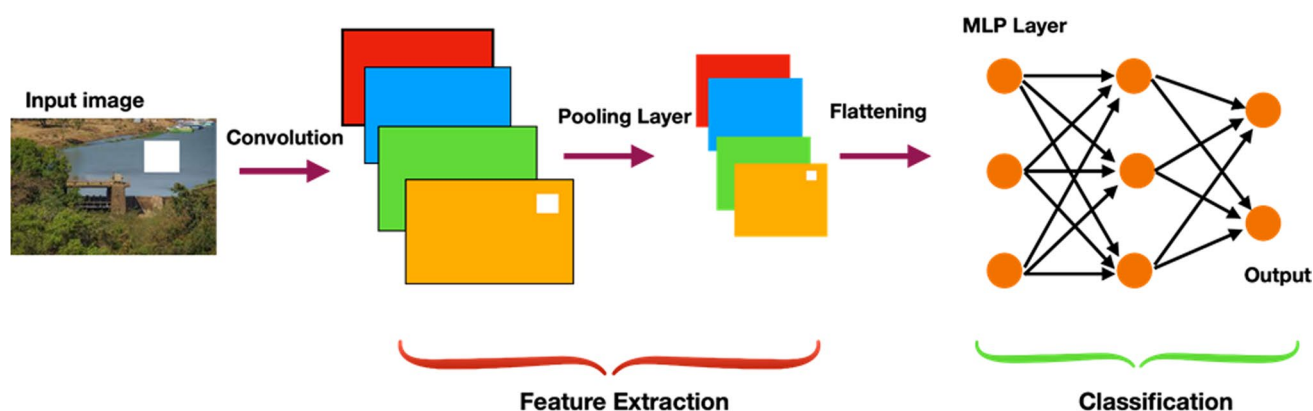


Fig. 6 Overview of CNN prediction scheme that involves hierarchical feature extraction through convolutional layers, followed by pooling to reduce dimensionality, fully connected (Multilayer Perceptron, MLP) layers then interpret the extracted features for final predictions

diseases, with an extensive overview of 19 studies in this domain. Additionally, Gikunda and Jouandeau [98] underscores the necessity for cutting-edge tools in agriculture, proposing a classification taxonomy for CNNs specifically tailored to agricultural applications [97]. Pre-trained CNN architectures like DenseNet-121, ResNet-50, VGG-16, and Inception V4 have shown impressive performance when fine-tuned for plant disease identification [99]. Additionally, CNNs have been extended to handle video data, demonstrating effectiveness in large-scale video classification [98]. As research continues to advance, CNNs will likely continue to drive innovation in agricultural technology.

2.2.2 Recurrent Neural Networks (RNNs)

RNNs are a type of neural network architecture designed to identify patterns in sequential data, such as language, speech, genomic sequences, and time series. They differ from feedforward networks due to their capability to retain information throughout the processing of a sequence via feedback connections (Fig. 7).

A range of studies have demonstrated the successful application of RNNs in agricultural tasks, particularly in the analysis of multitemporal remote sensing data for crop recognition and mapping. Bermudez et al. [100] compared RNN variants and found that Gated Recurrent Unit (GRU) achieved slightly better performance than LSTM. Castro et al. [101] proposed a hybrid RNN-CNN architecture for crop mapping, which outperformed alternative approaches. Gafurov et al. [102] further advanced the field by applying LSTM networks and spatial generalization in satellite data analysis, achieving a high accuracy of 93.7% in crop recognition. Pandey and Mishra [103] also highlighted the effectiveness of Artificial Neural Networks, specifically Generalized Regression Neural Network (GRNN), in predicting potato crop yield. These findings illustrate the effectiveness of RNNs in agricultural applications, particularly in handling temporal data for tasks

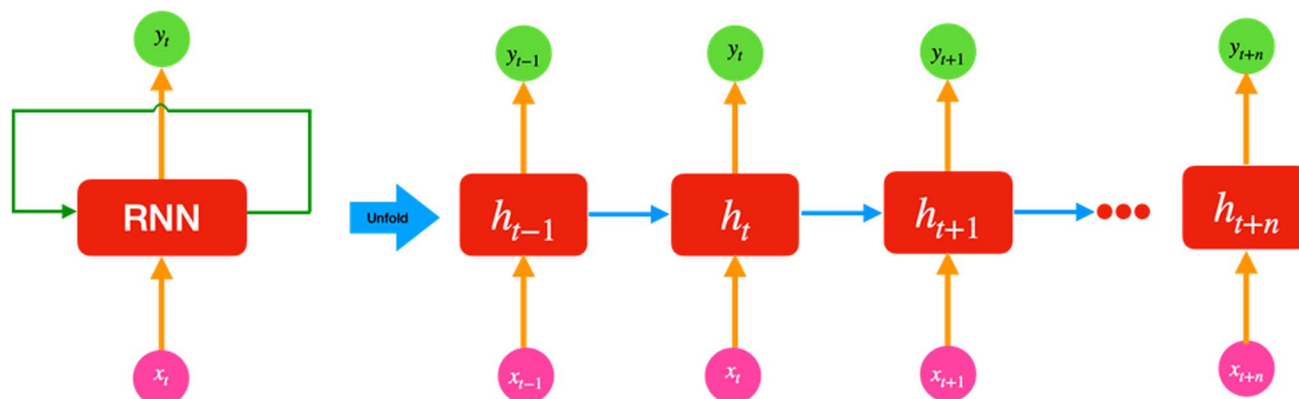


Fig. 7 Overview of RNN prediction scheme, each step in the sequence influences the next, allowing RNNs to capture temporal dependencies

such as crop recognition, mapping, and yield prediction. The ability of RNNs to model temporal context makes them valuable tools for leveraging multitemporal remote sensing data in agriculture.

2.2.3 Long Short-Term Memory (LSTMs)

LSTMs represent a specialized variant of RNNs that excel in capturing long-term dependencies within sequential data. Their distinctive architecture, featuring memory cells, enables them to retain and utilize relevant information over extended periods, making them particularly effective in tasks requiring the analysis of temporal relationships [104]. LSTMs have proven invaluable in various fields such as natural language processing, speech recognition, and environmental modeling, where understanding and predicting complex patterns over time is crucial [105, 106]. Their ability to mitigate issues like vanishing gradients commonly encountered in traditional RNNs, along with their flexibility in handling variable-length inputs and outputs, further solidifies their position as a powerful tool for tasks involving sequential data analysis (Fig. 8) [107].

The applications of LSTMs continue to expand across diverse domains, showcasing their versatility and effectiveness in addressing complex problems that involve temporal dependencies. A range of studies have demonstrated the potential of LSTM networks in various agricultural applications. Gafurov et al. [102] investigate accurate crop detection using satellite data, employing LSTM networks with a “spatial generalization” approach. LSTM networks, adept at learning long-term dependencies, outperform MLP and RF, achieving 93.7% accuracy. The study suggests that LSTM networks offer a promising solution for precise crop recognition in agriculture. Reuß et al. [108] also employed LSTM networks and RF algorithms to address the need for enhanced agricultural management for future food security. Their study assessed these algorithms’ performance for crop classification based on Sentinel-1 time series and meteorological data across Austria, the Netherlands, and France from 2015 to 2019. Results indicate that while both classifiers perform similarly for simple tasks, LSTM networks excel as scenarios become more complex, highlighting their value for large-scale applications in agriculture. Additionally, Wu et al. [109] proposed an outdoor agricultural robot that uses LSTM to predict the timing for watering and control resources. These studies collectively suggest that LSTM networks have significant potential to improve efficiency and accuracy in various agricultural tasks.

2.2.4 Generative Adversarial Networks (GANs)

GANs represent a category of DL algorithms capable of generating synthetic data resembling real data [110]. GANs comprise two neural networks: a generator network that produces synthetic data and a discriminator network that endeavors to distinguish between synthetic and real data (Fig. 9) [111]. The generator network is trained to generate synthetic data closely resembling real data, while the discriminator network aims to accurately discern whether the data is real or synthetic. Through an adversarial process, the two networks are trained concurrently until the generator generates synthetic data indistinguishable from real data [112].

In agriculture, GANs have been utilized for various tasks, such as synthesizing agricultural data to enhance limited datasets for training DL models or crafting realistic simulations for evaluating agricultural systems [113]. Lu et al. [114] provide a comprehensive review of GAN applications in agriculture, including plant health, weed detection,

Fig. 8 Overview of LSTM prediction scheme, that involves specialized memory cells that store and regulate information over extended time periods. These cells have gates (Forget, Input and Output) to control the flow of information

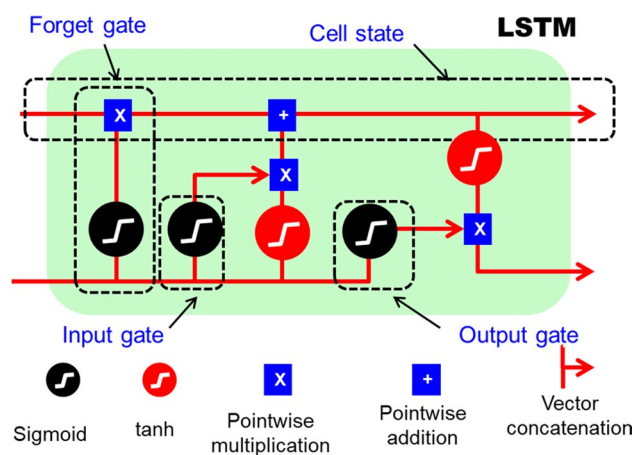
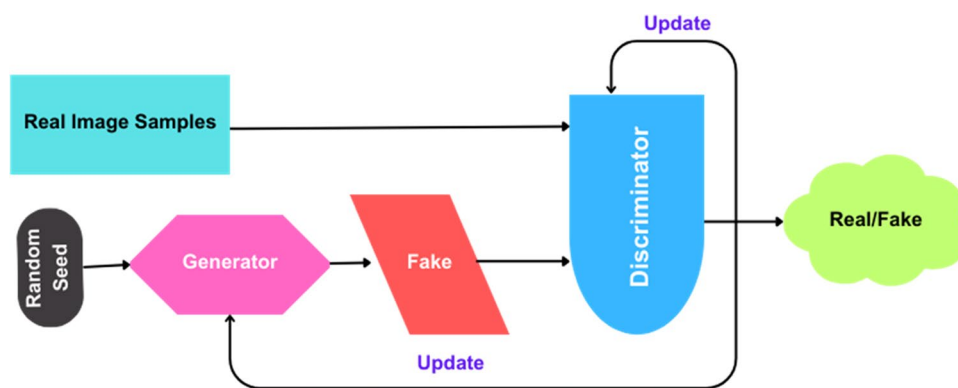


Fig. 9 Overview of GAN prediction scheme, where two neural networks, a generator and a discriminator, engage in a competitive process

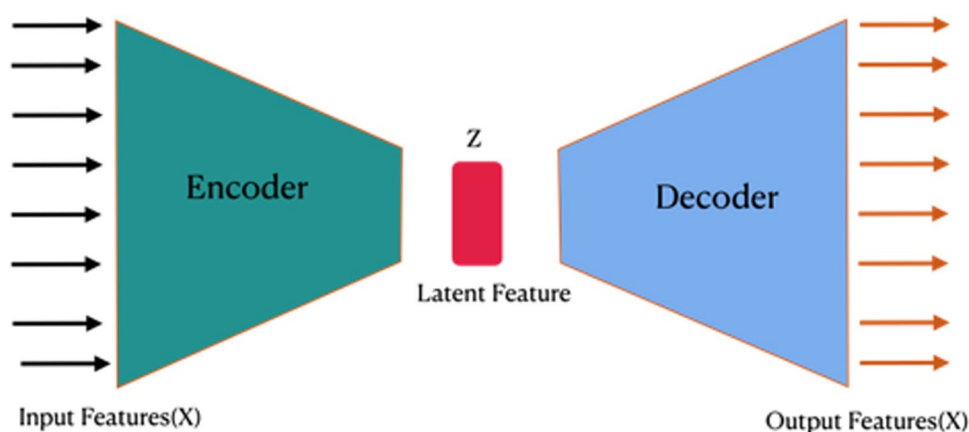


and postharvest fruit inspection. Fawakherji et al. [115] specifically focus on crop/weed segmentation in precision farming, using GANs to generate synthetic images for dataset augmentation. Zhao et al. [116] explore the use of pre-trained GANs for image generation in scenarios with limited data, demonstrating the potential for GANs to enhance training sets in agriculture. In general, GANs have shown promise in various agricultural applications, particularly in generating synthetic data to augment limited datasets for training DL models or creating realistic simulations for testing agricultural systems. As research continues to advance, GANs will likely continue to drive innovation in agricultural technology.

2.2.5 Autoencoders

It represents a class of DL algorithms comprising an encoder and a decoder, both tasked with acquiring a condensed representation of the input data and reconstructing the initial input (Fig. 10) [117]. They are often used for dimensionality reduction, feature learning, and anomaly detection. Autoencoders have been successfully applied in various agricultural applications, including crop health monitoring and anomaly detection [118]. These algorithms have been particularly effective in detecting subtle anomalies, outperforming linear Principal Component Analysis (PCA) in this task [118]. However, their performance can be affected by the presence of anomalies in the training data, leading to poor anomaly detection [119]. To address this, an Improved AutoEncoder for unsupervised Anomaly Detection (IAEAD) has been proposed, which manipulates the feature space to improve anomaly detection performance [119]. Autoencoders have also been used for feature learning in agricultural applications, with the ability to fulfill various purposes such as data visualization, denoising, and semantic hashing [120]. In the context of weakly supervised anomaly detection, a strategy has been proposed to transform input data into a more meaningful representation using autoencoders, leading to superior performance [121]. As research continues to advance, autoencoders will likely remain a powerful tool for agricultural applications, offering opportunities for dimensionality reduction, feature learning, and anomaly detection in diverse datasets.

Fig. 10 Overview of Autoencoders prediction scheme, involves an encoding phase where the input data is compressed into a lower-dimensional representation (encoder), followed by a decoding phase to reconstruct the original input (decoder)



2.2.6 Deep Belief Networks (DBNs)

DBNs are a type of DL algorithm consisting of multiple layers of stochastic restricted Boltzmann machines (RBMs) (Fig. 11). DBNs are primarily used for unsupervised learning tasks such as feature extraction and data clustering, which can be beneficial for tasks like crop classification and yield prediction in agriculture [122]. Key aspects of DBNs relevant to agricultural applications include: (1) DBNs can discover latent features in raw data without labeled information, which can then serve as input for supervised learning tasks, (2) DBNs can automatically extract informative features from raw data, which may improve the performance of downstream classification and regression tasks, and (3) it can group related samples, facilitating the exploration of underlying relationships among different classes [123].

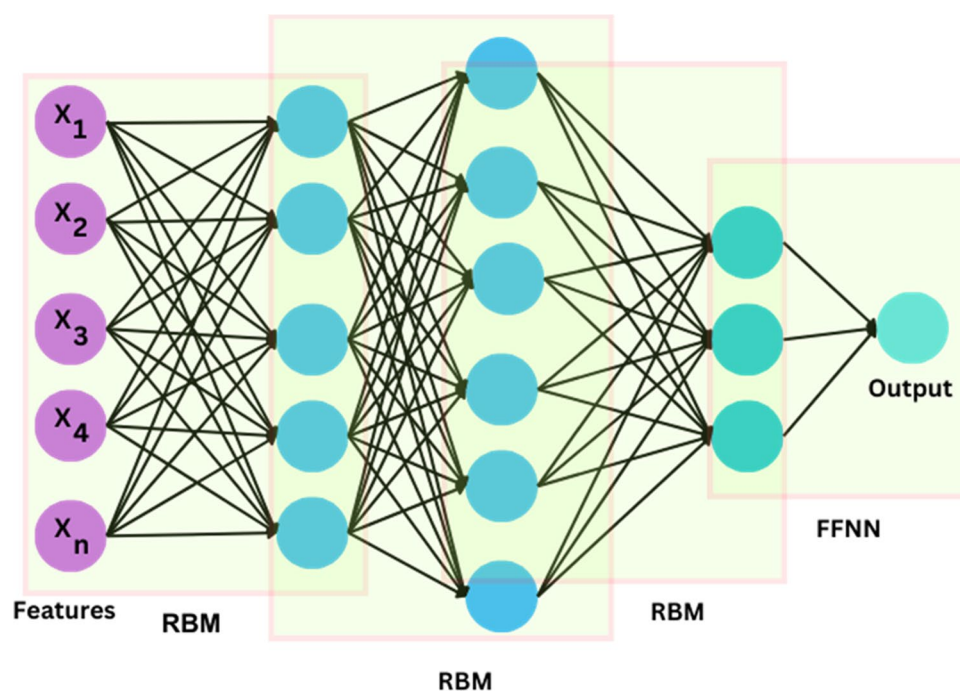
DBNs have also shown promise in various agricultural applications. Salakhutdinov and Murray [124] emphasized the capability of DBNs in clustering and classifying continuous data, by suggesting a technique for quantitative analysis of DBNs. While DBNs have not seen widespread adoption in recent years due to the rise of deep CNNs and GANs, they still hold relevance in particular scenarios requiring unsupervised learning capabilities [125].

2.2.7 Capsule Networks (CapsNets)

CapsNets are a relatively new DL architecture that aims to address some limitations of traditional CNNs, specifically regarding the representation of hierarchical relationships between features. Unlike CNNs, CapsNets represent each feature as a capsule containing a vector of weights, called the capsule vector, which captures the relationship between the feature and its instances (Fig. 12) [126]. Key aspects of CapsNets relevant to agricultural applications include: (1) CapsNets organize features into capsules, which contain vectors representing the relationships between features and their instances. This organization allows CapsNets to capture hierarchical relationships between features, which can be advantageous in agricultural applications where understanding the relationships between components is critical, (2) CapsNets utilize dynamic routing mechanisms to determine how much influence each lower-level capsule should exert upon higher-level capsules. This approach helps CapsNets learn the importance of individual features and their interactions and (3) it has been found to exhibit increased robustness to perturbations, such as noise, compared to traditional CNNs. This property can be beneficial in agricultural applications, where data quality might vary significantly [127–129].

Despite being less common than CNNs in agricultural applications, CapsNets have shown promise in certain scenarios, such as hyperspectral image classification and fabric defect detection [130]. The use of CapsNets in these scenarios is particularly beneficial due to their ability to understand the hierarchical relationships between features, which is crucial

Fig. 11 Overview of DBN prediction scheme, that involves unsupervised pre-training of each layer using Restricted Boltzmann Machines (RBMs) and subsequent fine-tuning with Feedforward Neural Network (FFNN)



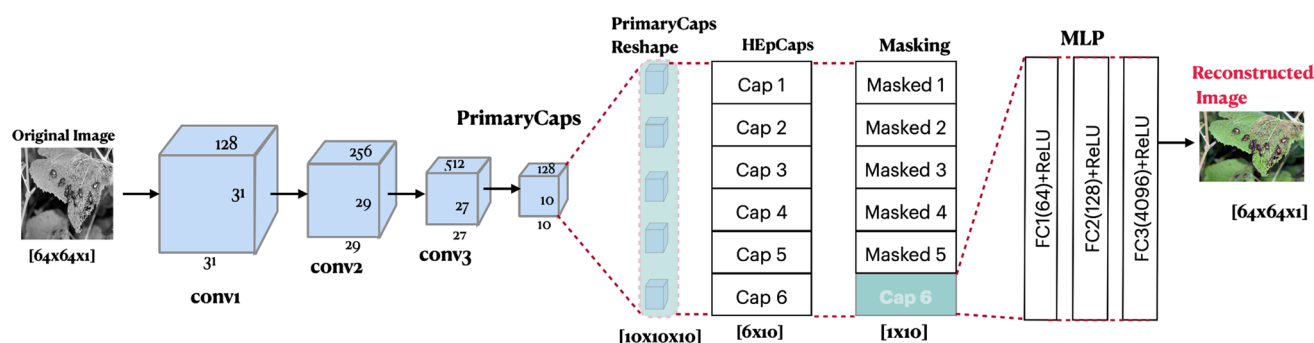


Fig. 12 Overview of CapsNets prediction scheme, that involves routing information capsules, each representing a specific entity or feature, through the network

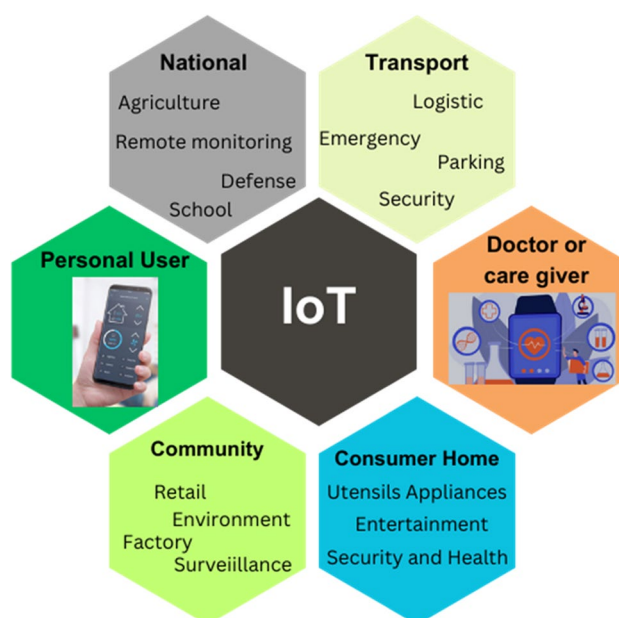
in these applications. As research continues to advance, CapsNets will likely see greater adoption in agricultural applications, particularly in cases where understanding the relationships between features is crucial.

2.3 The role of IoT in transforming agricultural systems

The IoT refers to a network of interconnected devices that are capable of collecting and exchanging data over the internet without requiring direct human intervention [131–133]. These devices, often embedded with sensors, software, and other technologies, can communicate with each other and with centralized systems to gather, analyze, and act upon data in various contexts. In simpler terms, IoT involves everyday objects, such as household appliances, vehicles, wearable devices, industrial equipment, and more, being equipped with internet connectivity and the ability to gather and share data. This connectivity allows these objects to interact with each other, exchange information, and perform tasks autonomously or with minimal human intervention. IoT technology enables a wide range of applications across different sectors, including agriculture, healthcare, transportation, manufacturing, smart homes, and cities (Fig. 13) [134, 135].

The IoT holds promise in revolutionizing agricultural systems by empowering farmers to gather and analyze data from diverse origins, such as sensors, drones, and satellites [136–139]. This technology aids farmers in making better-informed decisions regarding crop management, irrigation, and resource allocation, ultimately enhancing productivity and efficiency. Numerous studies have underscored the advantageous potential of IoT in agriculture. For instance, Rajeswar et al. [140] highlight the synergy between cloud-based big data analytics and IoT in advancing smart agriculture. IoT generates vast data necessitating cloud storage and analysis. This integration optimizes crop management, aiding decision-making

Fig. 13 The schematic of IoT application domains such as Personal, Community, Consumer Home, Doctor or care giver, Transport and National level users as per the data caption



and enhancing agricultural productivity. Additionally, Sreekantha and Kavya [141] highlight the transformative impact of the IoT on agriculture, providing farmers with various methods such as precision and sustainable farming. Their research reveals how IoT enables data collection on weather, soil conditions, crop monitoring, and pest detection, facilitating remote farm management through wireless sensor networks, microcontrollers, and smartphones. Ultimately, this technology helps reduce costs and enhance productivity in traditional farming practices [141].

Furthermore, IoT technology can facilitate precision agriculture, which employs data analytics and other tools to optimize crop yields while minimizing inputs like water, fertilizers, and pesticides. A study on IoT-based crop monitoring systems suggested utilizing light-dependent resistors, temperature sensors, and soil moisture sensors for gardening and agricultural management [142, 143]. Nonetheless, despite the potential advantages of IoT in agriculture, there exist challenges that must be addressed, including the high cost of IoT devices, the necessity for reliable connectivity in rural regions, and the absence of standardization in IoT technologies. In essence, while IoT can revolutionize agricultural systems by enabling data collection and analysis from diverse sources, challenges remain to be tackled to fully realize its potential (Fig. 14). As research progresses, IoT is expected to continue driving innovation in agricultural technology [144, 145].

3 Applications of ML, DL, and IoT in agriculture

The applications of ML, DL, and IoT in agriculture encompass a wide range of transformative possibilities [146, 147]. An illustration of rich and diverse ML, DL, and IoT applications for agriculture is provided in Fig. 15, showcasing their potential impact on various aspects of farming, from crop monitoring and disease detection to soil management, weather forecasting, and post-harvesting solutions. These technologies hold the potential to revolutionize the agricultural sector by optimizing resource use, enhancing productivity, and promoting sustainable practices.

3.1 Crop monitoring and yield prediction

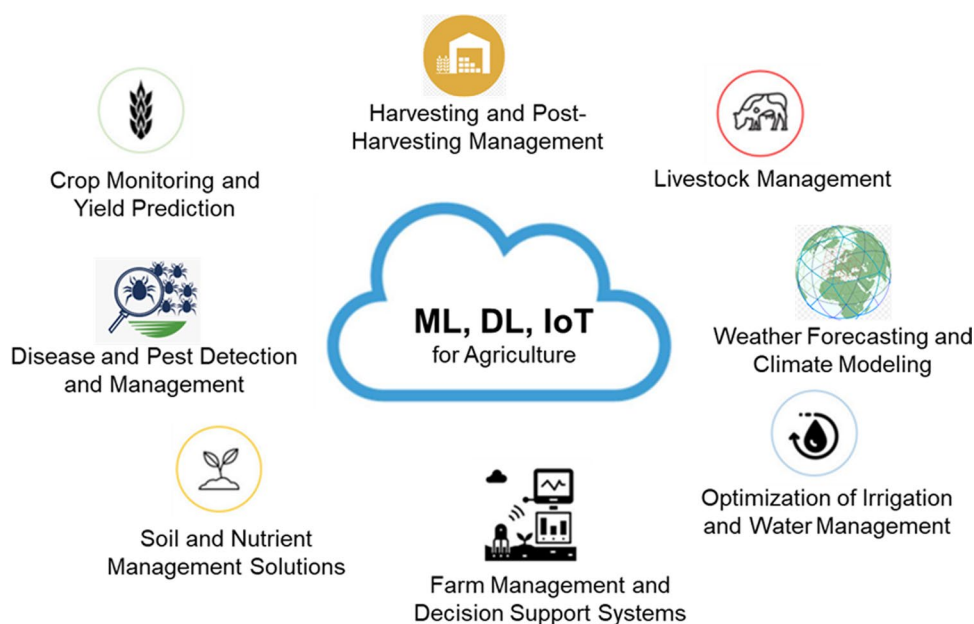
Crop monitoring and yield prediction are essential components of modern agriculture, empowering farmers to make educated choices, streamline resource distribution, and enhance overall crop output. The integration of ML, DL, and the IoT has revolutionized these aspects of agriculture, providing data-driven insights and real-time monitoring capabilities. For instance, the use of satellite-derived vegetation indices and stepwise linear regression has been successful in predicting crop yield under soil salinity, with a mean percent error of less than 10% [148]. Similarly, the integration of multi-rotor Unmanned Aerial Vehicles (UAVs), remote sensing, and ML methods has shown promise in monitoring wheat growth and predicting yield, with a relative root mean square error of 0.1030 [149]. Precision agriculture, which leverages sensing technologies and decision support models, has been identified as a key approach for crop yield prediction, with the potential for further enhancement through the use of artificial intelligence [150].

Rodriguez-Sanchez et al. [151] developed a data processing pipeline using ML to predict cotton yield from aerial imagery. Their Support Vector Machine classifier achieved 89% accuracy in identifying cotton pixels. Morphological

Fig. 14 The transformative impact of IoT on the agricultural sector by integrating smart sensors and devices, enabling real-time data collection and analysis



Fig. 15 In the agricultural sector, ML, DL, and IoT find practical applications. These include monitoring and predicting crop yields, detecting and managing diseases and pests, implementing solutions for soil and nutrient management, utilizing farm management and decision support systems, optimizing irrigation and water management, forecasting weather and climate, managing livestock, and overseeing harvesting and post-harvesting processes



processing and clustering predicted boll counts with an R^2 value of 0.93, demonstrating reliable pre-harvest yield estimation. Ikram et al. [34] introduced Smart Crop Selection (SCS), leveraging IoT and ML to address subpar crop yields due to inadequate soil analysis. SCS significantly enhances crop selection precision by integrating meteorological and soil data, achieving prediction accuracies of 97% to 98% across 11 crops with algorithms including DT, SVM, KNN, RF, and Gaussian Naïve Bayes. Historical rainfall data and Multiple Linear Regression further improve SCS's performance, with an exceptionally low Root Mean Square Error of 0.3% for rainfall prediction. This model surpasses traditional methods and existing IoT systems, providing comprehensive factor-driven predictions. Table 1 enlists diverse attributes and predictive techniques used to forecast crop yields within agricultural settings.

These examples display varied attributes and prediction methods employed within agronomy for estimating crop yields with precision. Evaluation criteria verify the efficacy of these techniques in accurately anticipating crop production levels. With ongoing progress in ML, DL, and IoT in agriculture, it's anticipated that their capabilities will expand, resulting in enhanced accuracy in crop yield predictions and increased overall agricultural productivity.

3.2 Disease and pest detection and management

Disease and pest detection and management are critical components of modern agriculture, ensuring food security and maintaining crop productivity. The integration of ML, DL, and IoT has revolutionized these aspects of agriculture, providing data-driven insights and real-time monitoring capabilities. Recent research and applications demonstrate the potential of these technologies in providing accurate detections, timely interventions, and enhanced overall productivity. For instance, a study by Johannes et al. [158] demonstrated the potential of these technologies to achieve high accuracy in disease detection. Another investigation conducted by Ngugi et al. [159] further emphasized the importance of early detection and the use of image processing and DL techniques in this context. Turkoglu and Hanbay [160] expanded on this by evaluating nine deep neural network architectures for plant disease detection. Utilizing DL and transfer learning methods, they assessed performance using real disease and pest images from Turkey. Deep feature extraction combined with SVM/Extreme Learning Method (ELM) classification outperformed transfer learning, achieving higher accuracy levels. These examples demonstrate the effectiveness of various feature types and prediction algorithms in accurately identifying diseases and pests in agriculture. As evidenced by the performance metrics, these approaches hold promise for enhancing disease and pest management strategies. As research in the field advances, the integration of ML, DL, and IoT technologies is expected to lead to further improvements in disease and pest detection and management, ultimately contributing to enhanced agricultural productivity. Table 2 displays a range of features and prediction algorithms employed in the detection and management of diseases and pests in agriculture.

Table 1 Various characteristics and forecasting methods for predicting crop yields

Aim	Inputs	Forecasting Algorithm	Performance	Refs.
Forecasting corn yields in US Corn Belt states	Temperature, precipitation, humidity, and solar radiation during the growing season	Ensemble and average ensemble models, alongside stacked LASSO models	Ensemble model outperforms individual models (RSME = 9.5%)	[152]
Forecasting Corn and Soybean Yields	Environmental data and management practices including crop genotype, weather conditions, soil conditions, and management practices	Hybrid CNN-RNN	RMSE = 9% (corn yield) and 8% (soybean yield)	[153]
Wheat yield prediction	Time-series NDVI data, meteorological variables, and yield data	RFR	Strong performance for regional model ($R^2 = 0.86$, $RMSE = 0.18 \text{ t ha}^{-1}$)	[154]
Rice yield prediction	long-term yield and weather datasets	Simple Multiple Linear Regression (SMLR) and ANN	ANN (deviations: -37.1% to 21.3% and a mean deviation of 4%), SMLR (deviations from -2.5% to 35.0% with a mean deviation of 16%)	[155]
Tomato yield prediction	Multi-temporal features from UAS data such as canopy cover, height, volume, Excessive Greenness Index, along with wea	ANN	Strong performance	[156]
Grape yield prediction	Images taken by smartphones capturing grape clusters, utilized for estimating grape weight	CNN	Accurately estimation of grape weight	[157]

Table 2 Different features and Prediction algorithms for disease and pest detection and management

Aim	Feature	Prediction algorithm	Performance	Refs.
Plant disease detection	Images of diseased and healthy plant leaves	CNN	Accuracy = 88.8%	[161]
Plant disease detection	Digital images of plant leaves	Transfer learning using CNN as feature extractor and SVM for classification	Accuracy = 88.77%	[162]
To detect plant diseases	Images of plant leaves for disease identification	Image segmentation with K-means clustering, HSV dependent classification, and feature extraction using GLCM	Accuracy = 98%	[163]
To develop a plant disease detection system	Data from IoT sensors monitoring plant health, combined with ML and DL algorithms	Three-stage model combining IoT, ML, and DL	Successfully predicted diseases in three classes of tomato	[164]
Plant leaf disease detection	Kaggle datasets of potato and tomato leaves for symptom investigation	Image processing and CNN (AlexNet and ResNet-50)	Achieved 97% and 96.1% accuracy with ResNet-50, and 96.5% and 95.3% accuracy with AlexNet for healthy-unhealthy leaf and leaf disease classification, respectively	[165]

3.3 Soil and nutrient management solutions

The convergence of IoT, DL, and ML technologies has transformed soil and nutrient management in agriculture, facilitating real-time monitoring and data-driven insights [166, 167]. Hossain et al. [168] propose an IoT-based soil nutrient monitoring and ML-driven crop recommendation system to enhance agricultural yield. By continuously sensing soil parameters and analyzing data through ML algorithms, the system offers tailored crop recommendations, reducing fertilizer costs and manual labor while boosting productivity for farmers and national growth. Combining IoT and Geographic Information System (GIS) enables continuous monitoring of soil health and nutrient levels, facilitating informed decisions on water and fertilizer usage [166]. Portable sensors like optical and electromagnetic sensors are suggested for efficient soil nutrient monitoring, promoting environmental responsibility [169]. ML and IoT applications are recognized for their potential in crop monitoring, disease detection, and water management, with ML algorithms aiding in forecasting and decision-making [167]. Additionally, IoT is proposed for monitoring soil parameters and crop productivity, utilizing sensors and cloud servers for data storage and farmer-centric application development [170]. These examples illustrate the diverse applications of ML, DL, and IoT in soil and nutrient management solutions in agriculture. Table 3 outlines the various characteristics and prediction algorithms utilized in soil and nutrient management solutions.

3.4 Optimization of irrigation and water management

The use of IoT sensors, ML, and DL algorithms in precision irrigation is a key advancement in agriculture, allowing for tailored irrigation and water conservation [175]. These technologies enable real-time data analysis and dynamic adjustments, optimizing irrigation schedules and reducing water wastage. The integration of cloud computing and IoT in smart irrigation systems further enhances their efficiency and accessibility [176]. The application of IoT in smart irrigation systems provides real-time data for environmental monitoring, improving production and product quality [177].

In terms of water resource allocation, ML algorithms optimize water distribution and balance competing demands [178], while DL techniques monitor crop health and detect water stress using satellite imagery and remote sensing data [178]. This integration, combined with IoT sensor feedback, is providing farmers with informed insights for judicious irrigation management [178]. The use of AI in water management is also highlighted, with a focus on efficient water treatment plants and water resource optimization [179]. Furthermore, the incorporation of adaptive decision support systems into precision irrigation management is enhancing water use efficiency and crop yields. Lastly, an IoT-based ML technique is being used to forecast water requirements in irrigation fields, aiding in the sustainable use of water resources [180]. Table 4 outlines the various characteristics and prediction algorithms utilized irrigation and water management.

3.5 Weather forecasting and climate modeling

Agriculture is significantly impacted by weather forecasting and climate modeling, influencing aspects such as crop growth, development, and yields, as well as the prevalence of pests and diseases [187, 188]. These forecasts are essential for both long-term strategic decisions and short-term tactical decisions in agriculture. Enhancing the value of forecasts can be achieved by integrating seasonal climate prediction with agricultural models. This integration offers valuable insights into agricultural practices and helps bridge the gap between farmers' needs and the available forecast information [189, 190].

Weather forecasting and climate modeling is a complex and evolving field, with various approaches being explored. Recent research indicates that data-driven approaches leveraging ML and DL can directly address weather and climate forecasting or projection tasks by learning data-driven functional mappings through deep neural networks [191]. These approaches have been applied to predict maximum temperature, rainfall, evaporation, and wind speed, with promising results [192]. However, the generality of these models is a concern, as they are trained on specific spatiotemporal tasks and lack the broad applicability of numerical models [191]. Despite these limitations, the use of ML and DL in weather forecasting and climate modeling has significant potential in agriculture, particularly in predicting drought conditions and assessing water stress levels [191].

Furthermore, IoT sensors and ML algorithms are being utilized to predict drought conditions and assess water stress levels in agricultural regions, enabling farmers to deploy drought-resistant crop varieties and adaptive irrigation strategies [193]. Additionally, these technologies have been used to create a prototype drought monitoring

Table 3 Different features and prediction algorithms for soil and nutrient management solutions

Aim	Feature		Prediction algorithm	Performance	Refs.
soil nutrient monitoring and crop recommendation system	soil nutrient data		IoT + ML	strong performance	[168]
soil properties affecting crop growth	Soil properties include organic matter, plant nutrients, and micro-nutrients		Back Propagation Neural Network (BPN)	BPN outperforms multivariate regression	[171]
Enhance soil nutrient and resource	Images from remote sensing for analysis, featuring vegetation characteristics		CNN	Accuracy = 99.58%	[172]
Enhance crop yield and nitrogen management	Surface temperatures, moisture levels, nitrogen availability data		CNN	high accuracy (99% for soil nutrient deficiency)	[173]
Analyze soil nutrient status	Soil features include boron, organic carbon, potassium, phosphorus, available boron, and pH		Extreme Learning Method (ELM) with various activation functions	strong Performance	[174]

Table 4 Different features and Prediction algorithms for optimizing irrigation and water management

Aim	Feature	Prediction algorithm	Performance	Refs.
Develop smart irrigation management system to predict field irrigation requirements	Utilizes sensors for soil and environmental data, integrates weather forecast for intelligent irrigation prediction	Smart algorithm	Strong performance	[181]
Develop automatic monitoring and control of irrigation processes	Data from various sensor devices including soil water content	Regression algorithm + IoT	Strong performance	[182]
Develop a smart irrigation system	Environmental parameters, weather forecasts, real-time data	Ensemble method + IoT	Accuracy = 90%	[183]
Develop an automated watering system	Sensor data including soil moisture levels	KNN + IoT	Strong performance	[184]
Develop an automated irrigation system	Sensor data including temperature and soil moisture levels	Application controlled monitoring system	Strong performance	[185]
Designing an IoT and smart irrigation system	IoT-based sensors for soil moisture, humidity, temperature, and light	Artificial algae algorithm (AAA) with least squares-SVM	Accuracy = 97.5%	[186]

system that combines satellite and climatic data. Furthermore, the application of artificial intelligence models, such as neuro-fuzzy adaptive inference systems, for forecasting meteorological drought has shown promising results [194]. In general, the integration of ML, DL, and IoT technologies offers robust tools for weather forecasting and climate modeling in agriculture. By leveraging data analytics, real-time monitoring, and adaptive control systems, farmers can enhance water use efficiency, increase crop yields, and advocate for sustainable water management practices. This empowers them to navigate the challenges posed by escalating water scarcity and climate variability with resilience and foresight. Table 5 presents different features and predictive techniques employed in weather forecasting and climate modeling.

3.6 Livestock management with ML, DL, and IoT

ML, DL, and the IoT are transforming livestock management through innovative solutions that enhance animal welfare, optimize farm operations, and improve productivity (Fig. 16). These technologies allow for real-time monitoring of livestock health, behavior, and environmental conditions, enabling farmers to make informed decisions based on data rather than intuition alone.

Recent studies have highlighted the potential of IoT and ML in predicting and monitoring livestock health. Kaur and Kaur [199] and Chaudhry et al. [200] both emphasize the importance of early detection and prevention of diseases in cattle, with Kaur and Kaur [199] specifically focusing on the use of IoT and ML to predict cattle health status. Chaudhry et al. [200] propose a real-time system for livestock health monitoring, incorporating a multi-sensor board and ML models for data analytics. Lee [201] extends this application to include the prediction of calving dates in breeding cows, using IoT and ML to monitor livestock activity and detect atypical symptoms. Various studies have also investigated the implementation of IoT and DL technologies for monitoring and managing livestock. For instance, Shinde and Prasad [202] introduced a framework utilizing IoT devices to gather sensor data, which was then analyzed through DL methods to identify sick or stressed animals in real time. Additionally, Kumar [37] developed a system that combined IoT and DL for livestock management, focusing on early detection of ill or distressed animals. By leveraging ML, DL, and IoT technologies, livestock managers can make data-driven decisions, improve animal care practices, and enhance overall farm efficiency and profitability. Table 6 shows various features and uses of predictive approaches in the area of animal management.

3.7 Harvesting and post-harvesting management solutions

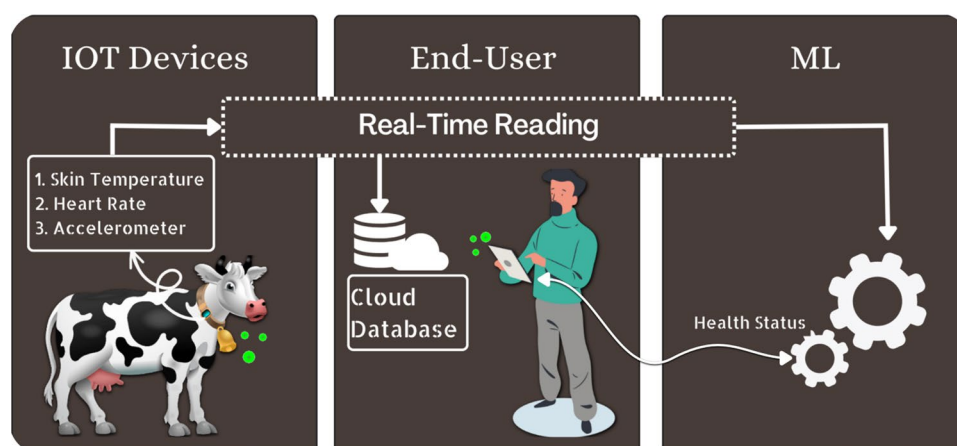
The integration of ML, DL, and the IoT is transforming harvesting and post-harvesting management solutions in agriculture. These technologies offer novel approaches to optimize harvesting processes, mitigate post-harvest losses, and enhance the quality of agricultural products [210]. ML algorithms, leveraging data from IoT sensors, satellite imagery, and weather forecasts, enable precise scheduling of harvest times based on crop maturity, weather conditions, and market demand, thus maximizing yield and quality while minimizing costs [211]. DL techniques play a crucial role in inspecting harvested crops for defects and ripeness using images from drones or IoT cameras, ensuring only high-quality produce reaches the market, reducing waste, and improving profitability [212]. Additionally, ML algorithms monitor storage conditions in warehouses and cold storage facilities, preserving product freshness and preventing spoilage by analyzing environmental data from IoT sensors [211]. Moreover, DL models optimize supply chain logistics and transportation routes, facilitating the timely delivery of agricultural products to markets while minimizing costs [212]. ML algorithms also analyze market data and pricing trends to inform marketing strategies and pricing decisions, allowing farmers and agribusinesses to maximize profits and mitigate financial risks.

Researchers have demonstrated the transformative potential of ML, DL, and IoT technologies in revolutionizing harvesting and post-harvesting management solutions in agriculture, enhancing efficiency, reducing waste, and ensuring product quality throughout the supply chain. For instance, Arthy et al. [213] have demonstrated how ML algorithms, combined with data from IoT sensors, satellite imagery, and weather forecasts, optimize harvesting schedules by predicting optimal harvest times based on crop maturity, weather conditions, and market demand. DL improves accuracy, reduces time, and enhances supply chain efficiency. These scholars' work highlights the transformative potential of ML, DL, and IoT technologies in revolutionizing harvesting and post-harvesting management solutions in agriculture, enhancing efficiency, reducing waste, and ensuring product quality throughout the supply chain.

Table 5 Different features and prediction algorithms used in agricultural weather forecasting and climate modeling

Aim	Feature	Prediction algorithm	Performance	Refs.
Weather Prediction	Climatic Characteristics	Linear Regressor, Polynomial Regressor, RF Regressor, DT Regressor, Linear Ridge Regressor, Linear Lasso Regressor, Linear ElasticNet Regressor, Linear BayesianRidge Regressor, SVR, and ANN regressor	RFregressors and DT regressors give better performance compared to others	[195]
Agricultural Drought Forecasting	Precipitation, Surface Pressure, Humidity, Temperature, Wind Speed, and Soil data	Prophet, VAR (Vector Auto-Regression), LSTM	LSTM model performed better than VAR and Prophet models	[196]
Rainfall Prediction	Past Weather Data	SVM and XGB	Strong performance	[197]
Weather Forecasting	Weather Data	Hybrid Decomposition Methods and Adaptive DL Strategy	Strong performance	[198]

Fig. 16 Integration of IoT and ML transforming livestock management



4 Challenges and possibilities in implementing ML, DL, and IoT in agriculture

In the realm of agriculture, the integration of ML, DL, and the IoT presents both challenges and prospects.

4.1 Challenges

Agricultural data can vary widely, be noisy, and frequently scarce, presenting a challenge in ensuring [214] the availability of high-quality data for training ML and DL models. This challenge is particularly pronounced in regions with limited connectivity or technological infrastructure [46]. Interpretable AI models and transparent decision-making processes are needed to build trust among stakeholders. Rural areas, where agriculture is predominantly practiced, often lack robust internet connectivity and technological infrastructure necessary for deploying IoT devices and accessing cloud-based ML/DL services [215]. Many farmers may lack the awareness, technical skills, and resources required to adopt and integrate ML, DL, and IoT technologies into their farming practices. Education and training programs are needed to empower farmers and facilitate technology adoption [216]. Collecting and sharing agricultural data, particularly sensitive information related to farm operations and yields, raises concerns about privacy and cybersecurity. Establishing robust data privacy protocols and cybersecurity measures is essential to address these concerns. Implementing ML, DL, and IoT solutions can be costly, particularly for small-scale farmers with limited financial resources. Ensuring affordability and demonstrating a clear return on investment (ROI) are essential to drive widespread adoption [214].

Expanding on these challenges in the context of Ethiopian agriculture, several specific issues arise that necessitate a closer examination and tailored solutions. Ethiopia, with its predominantly agrarian economy, faces unique hurdles in integrating advanced technologies into its agricultural practices. Firstly, the variability and scarcity of agricultural data in Ethiopia are exacerbated by the fragmented nature of smallholder farms, which constitute the majority of the agricultural sector. These small-scale farms often operate in isolation, making data collection inconsistent and incomplete. To address this, a concerted effort is needed to establish local data collection networks, perhaps leveraging community-based approaches where farmers collaborate to share data. Additionally, the use of satellite imagery and remote sensing technology can help fill data gaps and provide comprehensive insights into agricultural patterns and productivity.

The lack of robust internet connectivity and technological infrastructure in rural Ethiopia poses a significant barrier to the deployment of IoT devices and the use of cloud-based ML/DL services. Innovative solutions such as offline-capable technologies and edge computing can mitigate these challenges. For instance, deploying localized data processing units that do not require continuous internet access can enable farmers to benefit from real-time data analytics and decision-making tools. Another critical issue is the low level of awareness, technical skills, and resources among Ethiopian farmers. This gap can be bridged through targeted education and training programs that not only teach the basics of ML, DL, and IoT technologies but also provide practical, hands-on experience. Government and non-governmental organizations should collaborate to develop curricula that are accessible and relevant to local farmers' needs. Moreover, establishing demonstration farms where farmers can see these technologies in action can help build trust and showcase the tangible benefits of technology adoption.

Table 6 ML, DL, and IoT applications in animal management domain

Aim	Feature	Prediction algorithm	Performance	Refs.
Classify sheep behaviors using inertial motion sensors	Incorporating features from various time window sizes significantly	SVM, RF, Linear Discriminant Analysis (LDA)	Improved accuracy, with RF yielding the best results	[203]
Identify abnormal behaviors in dairy cows affected by subacute ruminal acidosis (SARA)	Continuous monitoring data from ruminal bolus for pH levels and positioning system for cow activity within the barn	MLP, DT, KNN, LSTM	KNN demonstrated the most effective performance, successfully detecting 83% of SARA cases	[204]
Predict milk production trends	Data from 91 lactating cows equipped with automatic milking systems (AMS) and thermo-hygrometric sensors, including daily average temperature-humidity index (THI) and milk yield, along with the position of the day in the lactation curve	RF	The model achieved an average relative prediction error of about 18% of daily production per cow and only 2% of the total milk production	[205]
Automate the recognition and scoring of pig posture	Depth images, including features such as convex hull parameters, boundary characteristics, and depth distances of shoulder and hip	SVM, CNN	Accuracy = 94.6368%	[206]
Identify invasive insects in close proximity to beehives	Multi-modal data including 3-D trajectories and IR imagery, along with multiple evidence sources obtained from trajectories	SVM, DT, KNN, CNN, Ensemble learning	Classification accuracy = 97.1%	[207]
Fish detection in recirculating aquaculture systems	Data from four underwater sensors under different conditions	CNN, YOLO (v5)	Improved accuracy with larger datasets and image augmentation; sensor selection crucial; YOLOv5 faster than Faster R-CNN	[208]
Identify aflatoxin-poisoned broilers	Acceleration data and behavioral observations	SVM, KNN, DT, Gradient Boosting Trees (GBT), RF	Models accurately detect poisoned broilers (> 86% accuracy); GBDT best at distinguishing between Normal and AF50 broilers (97% accuracy)	[209]

Privacy and cybersecurity concerns are particularly pertinent in the Ethiopian context, where there is often limited awareness of these issues. It is crucial to establish robust data privacy protocols and cybersecurity measures tailored to the local context. This includes educating farmers about the importance of data privacy and how to protect their information, as well as developing secure platforms for data sharing that comply with local regulations and international best practices. The cost of implementing ML, DL, and IoT solutions is a major concern for small-scale farmers in Ethiopia, who often operate with narrow profit margins. Affordability is arguably the most significant challenge for the adoption of these technologies. Many smallholder farmers struggle to invest in basic farming inputs, making the prospect of investing in advanced technologies seem unattainable. To ensure affordability, several strategies can be employed:

1. **Subsidies and grants:** Government subsidies and grants can significantly lower the initial cost burden for farmers. Programs designed to support technology adoption in agriculture can prioritize smallholder farmers to ensure they benefit from these initiatives.
2. **Microloans and financing options:** Providing access to microloans and flexible financing options tailored to the agricultural sector can help farmers manage the upfront costs of technology adoption. Financial institutions and cooperatives can play a crucial role in offering these services.
3. **Public–private partnerships:** Collaboration between the government, private sector, and non-governmental organizations can facilitate the development and deployment of affordable technologies. Public–private partnerships can help subsidize costs, provide technical support, and ensure sustainable implementation.
4. **Cost-effective solutions:** Developing and promoting cost-effective, low-maintenance technological solutions tailored to the needs of smallholder farmers is essential. Innovations that minimize the need for expensive infrastructure and offer scalable options can make technology adoption more feasible.
5. **Community-based approaches:** Encouraging community-based technology adoption, where resources and costs are shared among a group of farmers, can reduce individual financial burdens. Cooperative models can enable collective investment in technology, allowing farmers to benefit from economies of scale.
6. **Return on investment demonstrations:** Conducting pilot projects and case studies that demonstrate a clear return on investment can help convince farmers of the long-term benefits of these technologies. Showcasing successful implementations can build confidence and encourage wider adoption.

Finally, the appropriateness of these technologies for local farmers must be carefully considered. Technologies should be adapted to the local context, taking into account factors such as the scale of farming operations, available resources, and cultural practices. Simple, user-friendly interfaces and low-maintenance solutions are more likely to be adopted by smallholder farmers. Furthermore, involving farmers in the development and customization of these technologies can ensure that they meet the specific needs and constraints of the local agricultural community.

4.2 Future perspective

The integration of ML, DL, and IoT technologies into agriculture presents a transformative outlook for the industry. This integration promises increased productivity, reduced costs, and improved resource management. Despite its potential benefits, challenges such as data privacy concerns, technical barriers, and the need for farmer awareness and training remain significant hurdles. However, with anticipated increased investments in research and development, these technologies are poised to become more accessible globally. As a result, they offer solutions to pressing issues like climate change and food insecurity, while also enabling the development of advanced predictive models and algorithms for more precise agricultural practices.

Moving forward, the prospects of applying ML, DL, and IoT technologies in agriculture show substantial promise. Precision agriculture stands to benefit greatly from the optimization of resource use and enhanced crop yields through targeted interventions based on real-time data and analytics (Fig. 17). Smart farming systems will emerge, automating tasks like irrigation and pest detection to increase efficiency and reduce labor. Additionally, the use of DL algorithms for crop monitoring and management allows for early disease detection and optimized environmental conditions, further improving agricultural practices.

ML-based predictive analytics can optimize supply chain management, minimizing waste and ensuring timely delivery of agricultural products. Sustainable agriculture practices can be promoted through the optimization of resource use and reduction of chemical inputs. Moreover, these technologies have the potential to empower small-scale farmers by providing access to decision-support tools and market information, thus enhancing productivity and

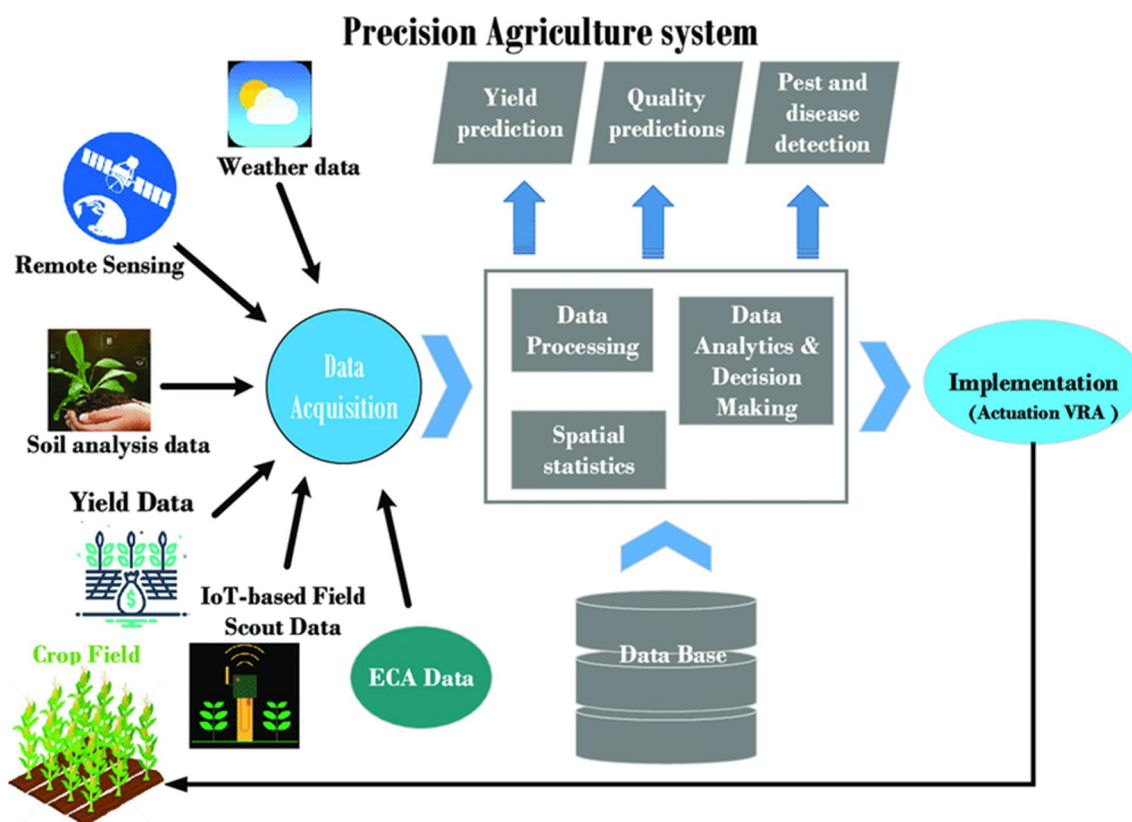


Fig. 17 Precision agriculture poised for advancements through real-time data and analytics

profitability. Furthermore, they offer resilience to climate change by providing early warnings for extreme weather events and optimizing water management.

However, realizing these benefits requires collaboration among farmers, technology developers, policymakers, and researchers. Overcoming barriers and embracing innovation are essential to harness the full potential of these technologies. By doing so, the agricultural sector can drive sustainable growth, improve food security, and mitigate the impact of climate change. It is crucial to address challenges such as data privacy, technical barriers, and farmer awareness to ensure successful implementation and widespread adoption of these technologies.

Expanding on the future perspective within the specific context of Ethiopian agriculture, it is essential to consider the unique challenges and opportunities that the country faces. Firstly, the issue of affordability is paramount. Most Ethiopian farmers are smallholders with limited financial resources, making the adoption of advanced technologies a significant challenge. Addressing affordability requires innovative solutions such as subsidized technology programs, affordable financing options like microloans, and community-based approaches where costs and resources are shared among groups of farmers. Demonstrating a clear return on investment through pilot projects and case studies can also help build trust and encourage adoption.

The lack of robust infrastructure in rural areas poses another significant barrier. Limited internet connectivity and the absence of reliable power sources can hinder the deployment of IoT devices and cloud-based ML/DL services. To overcome these limitations, the development of offline-capable technologies and edge computing solutions that can function independently of continuous internet access is crucial. Localized data processing units and solar-powered devices can provide practical solutions to infrastructure challenges. Farmer awareness and technical skills are also critical factors. Many Ethiopian farmers may be unfamiliar with the potential benefits and practical applications of ML, DL, and IoT technologies. Education and training programs tailored to the local context are essential to bridge this gap. Collaborations between government, NGOs, and private sector stakeholders can facilitate the development and dissemination of these programs, ensuring they are accessible and relevant to the needs of local farmers. Demonstration farms and hands-on training sessions can further enhance farmer engagement and confidence in adopting new technologies.

Data privacy and cybersecurity concerns must be addressed to build trust among farmers. Establishing robust data privacy protocols and educating farmers about the importance of data security are essential steps. Secure platforms for data sharing that comply with local regulations and international best practices can help alleviate these concerns. Furthermore, the appropriateness of these technologies for the Ethiopian context must be carefully considered. Solutions should be tailored to the specific needs and constraints of smallholder farmers. User-friendly interfaces, low-maintenance requirements, and scalable options that can grow with the farmers' needs are more likely to be adopted successfully. Involving farmers in the development and customization of these technologies can ensure they are relevant and practical for local agricultural practices.

Finally, collaboration among stakeholders is vital for the successful integration of ML, DL, and IoT technologies in Ethiopian agriculture. Policymakers, researchers, technology developers, and farmers must work together to create an enabling environment for technology adoption. This includes developing supportive policies, fostering innovation, and ensuring that the benefits of these technologies are equitably distributed.

In general, while the integration of ML, DL, and IoT technologies into Ethiopian agriculture presents significant challenges, it also offers transformative potential. By addressing affordability, infrastructure limitations, farmer awareness, data privacy, and the appropriateness of technology, it is possible to create a more resilient and technologically advanced agricultural sector in Ethiopia. Through collaboration and innovation, the agricultural sector can drive sustainable growth, improve food security, and contribute to mitigating global challenges such as climate change.

5 Conclusion

The integration of ML, DL, and IoT technologies presents a transformative opportunity for agriculture, particularly in developing countries like Ethiopia, which faces significant food security challenges. Leveraging data-driven insights through these technologies enables farmers to optimize resource allocation, enhance productivity, and mitigate risks related to crop management, soil health, and weather patterns. Real-time monitoring and control facilitated by IoT devices further contribute to sustainability and efficiency in agricultural operations. However, the adoption of these technologies in Ethiopia is accompanied by notable challenges. Data quality and connectivity limitations, particularly in rural areas, hinder the effective implementation of ML, DL, and IoT solutions. The need for farmer education and awareness is also critical, as many farmers lack the technical skills required to utilize these advanced technologies effectively. Affordability is a particularly pressing issue. Many smallholder farmers in Ethiopia operate with limited financial resources, making the high cost of technology a significant barrier to adoption. Addressing affordability requires targeted strategies, including government subsidies, microloans, and community-based approaches that can lower the initial financial burden on farmers. Additionally, demonstrating a clear return on investment through pilot projects and cost-effective solutions can help justify the investment in these technologies. To overcome these barriers, collaborative efforts are essential. Government, industry stakeholders, and research institutions must work together to invest in necessary infrastructure, develop supportive policies, and implement training programs tailored to the needs of Ethiopian farmers. This collaboration is crucial to ensure the successful integration of ML, DL, and IoT technologies in agriculture.

Despite these challenges, the potential benefits are substantial. By harnessing the power of ML, DL, and IoT, Ethiopia can improve food security, alleviate poverty, and stimulate economic growth. This review emphasizes the importance of understanding the current status, challenges, and future possibilities of these technologies in agriculture. It highlights their transformative potential in addressing critical agricultural issues and driving sustainable development in the sector. By understanding and addressing the challenges of technology adoption, stakeholders can unlock the full transformative potential of ML, DL, and IoT, paving the way for a more resilient and prosperous agricultural sector in Ethiopia.

Author contributions N.E.B., M.D.C, and A.G.S.: conceptualization, methodology, resources, software, visualization, writing—original draft, writing—review & editing. B.W.D and T.S: conceptualization, supervision, writing—review & editing. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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